ANDRE'S THEOREM: REGULARITY, FLATNESS WITH GEOMETRICALLY REGULAR FIBRE, AND FORMAL SMOOTHNESS OF A LOCAL MAP

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Equivalence of flatness with geometrically regular fibre, flatness with formally smooth fibre, and formal smoothness of the map. Proposition \ref{proposition-fs-flat-fibre-fs}, cited at proof lines 12141-12142. Gives the equivalence of the last three conditions.
- Regularity implies flatness with geometrically regular fibre. used at proof lines 12143-12144. Gives one of the implications.
- Andre's theorem on formal smoothness implying regularity. the reference environment citing \cite{Andre-smooth}, cited at proof lines 12119-12121. Supplies the substantive implication; the reference environment is a comment environment in this chapter.

Source reference. Original lemma 0H7U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21469–21469. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07GG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9516–9516. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9842–9842. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9770–9770. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9443–9443. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9802–9802. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07QI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9833–9833. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9773–9773. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-Andre-smooth: sources/stacks-more-algebra/pinned-original-source.tex, source lines 12118–12118. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Theorem 0.2 (André)**.** *Let $A \to B$ be a local homomorphism of Noetherian local rings. Let k be the residue field of A and $\overline{B} = B \otimes_A k$ the special fibre. Assume $A \to A^\wedge$ is regular*[1]*. The following are equivalent*

(1) *$A \to B$ is regular,*
(2) *$A \to B$ is flat and $\overline{B}$ is geometrically regular over k,*
(3) *$A \to B$ is flat and $k \to \overline{B}$ is formally smooth in the $\mathfrak{m}_{\overline{B}}$-adic topology, and*
(4) *$A \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology.*

Proof. We have seen the equivalence of (2), (3), and (4) in Proposition 07NQ. It is clear that (1) implies (2). Thus we assume (4) holds and we prove (1).

By Lemma 07ED we see that $A^\wedge \to B^\wedge$ is formally smooth. By Proposition 07PM we see that $A^\wedge \to B^\wedge$ is regular. By assumption $A \to A^\wedge$ is regular, hence $A \to B^\wedge$ is regular by Lemma 07QI. Since $B \to B^\wedge$ is faithfully flat, we conclude that $A \to B$ is regular by Lemma 07NT. □

[1]For example, if A is a G-ring or (quasi-)excellent, see Sections 07GG and 07QS.